

- The operator shall respond to operating signals from the proper signal person only but shall obey a stop signal at any time from anybody.
- Take action to prevent unauthorized personnel from entering or approaching the job site area.
- As an operator, you must always be able to see the point of work location. If this is not possible then a spotter must be used. If visibility should become blocked for any reason, stop operation immediately.

2.1.10 Wind

Wind can have a significant effect on loads that may be lifted by a crane. Wind forces act differently on a crane, depending upon the direction from which the wind is blowing (e.g., wind on the rear of the boom can result in decreased forward stability, wind on the underside of the boom can result in decreased backward stability, and wind on the side of the boom can result in structural damages, etc.).

Sany Group recommends that if the wind speed (velocity) is between 32 km/h to 48 km/h, that the load capacities be reduced to account for the size and shape of the load and the wind direction in relation to the machine for all boom, and swingaway boom extension lengths.

Sany Group also recommends that operation of the crane in wind velocities over 48 km/h be prohibited. To assist you in determining prevailing wind conditions.

Wind Velocities and Their Effects:

Wind force		Wind speed		Effect of the wind in the inland
Beaufort	Description	[m/s]	[km/h]	
Zero (0)	Calm	0-0.2	1	No wind, smoke rises straight up
1	Slight air (draft)	0.3-1.5	1-5	Wind direction is shown only by observing the trail of smoke, not by the wind sock
2	Light breeze	1.6-3.3	6-11	Wind can be felt on the face, the leaves rustle, wind sock moves slightly
3	Gentle breeze	3.4-5.4	12-19	Leaves and thin twigs move, wind extends a small breeze flag
4	Moderate breeze	5.5-7.9	20-28	Swirls up dust and loose paper, moves twigs and thin branches
Reduce crane load ratings and operating parameters at 32 km/h.				
5	Fresh breeze	8-10.7	29-38	Small deciduous trees begin to sway, foam forms at sea
6	Strong breeze	10.8-13.8	39-49	Thicker branches move; telephone lines begin to whistle, umbrellas are difficult to use
Cease all craning operations at 48 km/h; lower and retract boom.				

Wind force		Wind speed		Effect of the wind in the inland
Beaufort	Description	[m/s]	[km/h]	
7	Stiff wind	13.9-17.1	50-61	Entire trees swaying; difficult to walk into wind
8	Gale force wind	17.2-20.7	62-74	Breaks twigs off trees, walking becomes difficult
9	Gale	20.8-24.4	75-88	Minor damage to property (chimney tops and roofing tile are blown off)
10	Severe gale	24.5-28.4	89-102	Trees are uprooted, significant damage to property
11	Violent storm	28.5-32.6	103-117	Extensive, widespread storm damage
12	Hurricane	32.7-36.9	118-133	Major destruction

2.1.11 High Altitude

The maximum working altitude is 3500m for this crane.

If need this crane to work on a location over this working altitude, specify it in the contract. Sany provide optional design to make the crane working on a place up to 5000m.

The crane power performance is reduced by high altitude as below:

Altitude(m)	Reductions of crane power performance by every 1000 metres altitude increase
0-2500	2%-5%
2500-3500	8%-16%
> 3500	Sharply

2.1.12 Electrocution Hazard

Keep all parts of this machine away from all electrical power lines and other electrical power sources as shown on the following table:

Minimum Distances Between Machine and Power Lines

Cable voltage, kV (kilovolts)	Minimum required clearance, m
350 or less	6.10
Over 350 - 500	7.62
Over 500 - 750	10.67
Over 750 - 1,000	13.72